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THE PERFECT HAT-TRICK WITH ALL-INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023



BOTANY

1. Read the following statements and choose the correct option.

Statement-I: The word systematics is derived from the Latin word "*systema*" which means systematic arrangement of organisms.

Statement-II: Systematics takes into account evolutionary relationship between organisms.

(1) Both Statement-I and Statement-II are true.

(2) Statement-I is false but Statement-II is true.

(3) Statement-I is true but Statement-II is false.

(4) Both Statement-I and Statement-II are false.

2. Read the Assertion and Reason carefully and choose the correct option.

Assertion(A): Genus may have one or more than one specific epithet.

Reason(R): Genus comprise a group of related species.

(1) Both A and R are correct and R is the correct explanation of A.

(2) Both A and R are correct but R is not the correct explanation of A.

(3) A is not correct but R is correct.

(4) A is correct but R is not correct.

3. Match the following

Column-I		Column-II	
A	*Cuscuta*	i	Insectivorous plants
B	Bladderwort	ii	Imperfect fungi
C	*Trichoderma*	iii	Unicellular fungi
D	*Sacchromyces*	iv	Parasitic plant

(1) A(iv), B(i), C(ii), D(iii)

(2) A(i), B(ii), C(iv), D(iii)

(3) A(iii), B(iv), C(ii), D(i)

(4) A(ii), B(iii), C(iv), D(i)

4. Which one of the following is not a criterion for classification of fungi

(1) Mode of spore formation

(2) Type of fruiting body

(3) Morphology of mycelium

(4) Mode of nutrition

5. In gymnosperms, female gametophyte

(A) Act as an endosperm.

(B) Bears two or more archegonia.

(C) Is dependent on sporophyte .

Choose the correct option to complete the sentence.

(1) Only B is correct

(2) A and B are correct

(3) A and C are correct

(4) A, B and C are correct

6. *Ectocarpus* is a

(1) Thalloid branched filamentous red alga.

(2) Colonial brown alga.

(3) Branched filamentous brown alga.

(4) Filamentous branched green alga.

7. Identify the suitable answer.

(A) Potato is a modified root while sweet potato is a modified stem.

(B) Leaves are arranged in basipetal order on main stem or branches.

(C) China rose, mustard, sunflower shows alternate phyllotaxy.

(D) In the flowers of pea or bean the large posterior petal overlaps the two lateral petals.

(1) A and B correct, C and D incorrect.

(2) only A is incorrect but B, C and D are correct.

(3) A, B and C are correct but D incorrect.

(4) A and B incorrect but C and D are correct.

8. The character of flower represented by floral formula but not by floral diagram.

(1) Position of ovary

(2) Adhesion of stamens

(3) Placentation

(4) Aestivation

9. Which of the following statement is incorrect ?

(1) The mustard flower is tetradynamous

(2) The androecium of disc florets of *Helianthus* are epipetalous and syngenesious

(3) The ray florets of Asteraceae members have an inferior ovary

(4) China rose is a unisexual flower with an inferior ovary

10. Match the following columns

Column-I		Column-II	
1	Sieve tube elements	A	Help in maintaining pressure gradient in sieve tube
2	Companian cells	B	Stores food material
3	Phloem parenchyma	C	Sclerenchymatous cells
4	Phloem fibres	D	Enucleated and associated with companion cells

(1) 1 – D; 2 – A; 3 – B; 4 – C

(2) 1 – A; 2 – B; 3 – C; 4 – D

(3) 1 – B; 2 – C; 3 – D; 4 – A

(4) 1 – C; 2 – B; 3 – A; 4 – D

11. Lateral roots arises from

(1) Epiblema (2) Pericycle

(3) Endodermis (4) Cambium

12. Ribosomes are composed of

(1) DNA and RNA

(2) DNA and Proteins

(3) rRNA and Proteins

(4) rRNA and Phospholipids

13. **Assertion (A)** : In acrocentric chromosomes, the centromere is situated close to one end, forming one exteremely short arm and one very long arm

Reason (R): The position of the centromere determines the length of the arms of a chromosome, resulting in unequal arms in chromosomes like acrocentric ones.

Choose the correct option

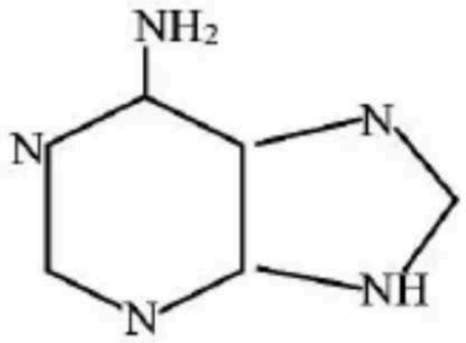
(1) Both A and R are correct, and R is the correct explanation of A

(2) Both A and R are correct, but R is not the correct explanation of A

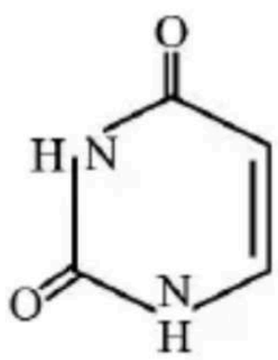
(3) A is incorrect, but R is correct

(4) A is correct, but R is incorrect

14. Identify the structures A and B



(A)



(B)

A	B
(1) Adenine	Cytosine
(2) Adenine	Uracil
(3) Guanine	Uracil
(4) Guanine	Cytosine

15. Select the incorrect match

(1) Trypsin – Enzyme

(2) Collagen – Intracellular ground substance

(3) Insulin – Hormone

(4) Antibody – Defence against infection

16. Read the following statements and choose the correct option.

Statement-I: During 'G_o' phase of cell cycle, the cell is metabolically inactive.

Statement-II: The centrosome undergoes duplication during S phase of interphase.

(1) Both Statement-I and Statement-II are true.

(2) Statement-I is false but Statement-II is true.

(3) Statement-I is true but Statement-II is false.

(4) Both Statement-I and Statement-II are false.

17. During meiosis, the haploid condition is achieved by the separation of sister chromatids in which of the following stages?

(1) Anaphase I

(2) Anaphase II

(3) Metaphase II

(4) Prophase II

18. C_4 plants differ from C_3 plants in which of the following aspects ?

a) The initial substrate that accepts CO_2

b) The type of end product formed during the C_3 cycle

c) The type of pigments involved in photosynthesis

d) The total number of ATP molecules consumed to fix 6 molecules of CO_2

(1) (a) and (d)

(2) (a) only

(3) (a) and (c)

(4) (a), (b), (c) and (d)

19. Chemiosmosis requires

I) a membrane II) A proton pump

III) Proton gradient IV) ATP synthase

(1) I and IV only

(2) I, II, and III only

(3) I, II and IV only

(4) I, II, III and IV

20. Assertion (A): The inner membrane of the mitochondrion contains systems involved in electron transport.

Reason (R): The mitochondrial matrix contains all the enzyme complexes of the Krebs cycle except succinate dehydrogenase.

Choose the correct option:

(1) Both A and R are correct, and R is the correct explanation of A.

(2) Both A and R are correct, but R is not the correct explanation of A.

(3) A is incorrect, but R is correct.

(4) A is correct, but R is incorrect

21. Redox equivalents or hydrogen generated during oxidation of succinate in the TCA cycle is received by

(1) Complex I (2) Complex II

(3) Complex III (4) Complex IV

22. Match the contents of Column I and Column II.

Column-I		Column-II	
A	Auxins	I	Coconut milk factor
B	Gibberellins	II	Apical hook formation
C	Cytokinins	III	Internodal elongation
D	Ethylene	IV	Apical dominance

(1) A – IV, B – I, C – II, D - III

(2) A – IV, B – III, C – I, D – II

(3) A – III, B – IV, C – I, D – II

(4) A – III, B – I, C – IV, D – III

23. Read the following statements and choose the correct option.

Statement-I: ABA promote fruit like apple to elongate and improve its shape.

Statement-II: Ethylene is largely a growth inhibitor.

(1) Both Statement-I and Statement-II are true.

(2) Statement-I is false but Statement-II is true.

(3) Statement-I is true but Statement-II is false.

(4) Both Statement-I and Statement-II are false.

24. The chemical nature of Kinetin is

(1) Indole 3-acetic acid.

(2) N^6 – furfurylaminopurine.

(3) Indole butyric acid.

(4) Naphthalene acetic acid.

25. Which of the following statements about the monosporic embryo sac is incorrect?

(1)Polygonum type embryo sac is monosporic, 7-celled, and 8-nucleated.

(2) During embryo sac formation 6 nuclei out of 8 nuclei go through cytokinesis or wall formation

(3) Polar nuclei are responsible for the formation of antipodal cells in the embryo sac.

(4) The egg apparatus is located at the micropylar end of the embryo sac.

26. **Assertion (A)**: The period for which pollen grains remain viable is highly variable.

Reason (R): Pollen viability is influenced by species-specific factors as well as environmental conditions like temperature and humidity.

Choose the correct option:

(1) Both A and R are correct, and R is the correct explanation of A.

(2) Both A and R are correct, but R is not the correct explanation of A.

(3) A is incorrect, but R is correct.

(4) A is correct, but R is incorrect.

27. Which of the following statement is incorrect?

(1)Apomixis is a form of asexual reproduction that mimics sexual reproduction.

(2) Seeds are always the products of pollination and fertilization and cannot be

formed by apomixis.

(3) The epiblast in a monocot embryo is considered a rudimentary cotyledon.

(4) Residual, persistent nucellus in a seed is known as perisperm

28. **Statement I**: In snapdragon, F1 plants exhibit neither red nor white flowers.

Statement II: Snapdragon flowers show incomplete dominance,where neither of the two alleles is completely dominant over the other, resulting in intermediate inheritance.

Choose the correct option:

(1)Both Statement I and Statement II are true.

(2)Statement I is false, but Statement II is true.

(3)Statement I is true, but Statement II is false.

(4)Both Statement I and Statement II are false

29. In f_2 generation of Mendelian dihybrid cross which of the following genotypes appear with a probability of 1/16?

(1)AABB, AaBB, AaBb

(2)AaBb, AABb, AAbb

(3)AABB, AABb, aaBB

(4)AABB, AAbb, aaBB, aabb

30. Match the contents of Column I and Column II.

Column-I		Column-II	
A	Regnald C. Punnet	I	Law of Inheritance
B	Mendel	II	Linkage and Recombination
C	Sutton and Boveri	III	Genotype probability
D	T.H. Morghan	IV	Parallel behaviour of genes and chromosomes.

(1) A – III, B – I, C – IV, D - II

(2) A – III B – IV, C – I, D – II

(3) A – II, B – III, C – IV, D – I

(4) A – I, B – II, C – III, D – IV

31. If the Meselson and Stahl experiment begins with $^{15}N/^{15}N$ DNA and is continued for four generations in bacteria growing in a medium containing only ^{14}N, what will be the ratio of $^{15}N/^{15}N$: $^{15}N/^{14}N$: $^{14}N/^{14}N$ DNA after the fourth generation?

(1) 1 : 1 : 0 (2) 1 : 4 : 0

(3) 0 : 1 : 3 (4) 0 : 1 : 7

32. In prokaryotes the initiation and termination factors of transcription are respectively

(1) σ and ρ (2) ρ and σ

(3) σ and β (4) ρ and τ

33. Read the following statements (a-e) and identify which are correct:

(a) A gene is the functional unit of inheritance and is located on DNA.

(b) Monocistronic structural genes are mostly found in eukaryotes, while polycistronic genes are mostly found in prokaryotes.

(c) Exons are coding sequences that appear in mature or processed RNA.

(d) Introns are non-coding sequences that do not appear in mature or processed

RNA.

(e) Promoter and regulatory sequences, though not coding for RNA or protein,can influence the inheritance of a character.

Options:

(1) (a), (b), (c) only

(2) (a), (c), (d), (e) only

(3) (a), (b), (d), (e) only

(4) All of the above (a, b, c, d, e)

34. *Monascus purpureus* is a source of

(1) Blood clot removing agent.

(2) Blood purifying agent.

(3) Blood cholesterol lowering agent.

(4) Immunosuppressive agent.

35. Match the Columns

Column-I		Column-II	
A	Citric acid	I	Inhibiting enzyme responsible for synthesis of cholesterol .
B	Statins	II	Used in organ transplantation
C	Streptokinase	III	Produced by *Aspergillus*
D	Cyclosporine	IV	Important agent for use in treatment during myocardial infarction (MI).

(1) A – I, B – II, C – III, D – IV

(2) A – III B – IV, C – II, D – I

(3) A – III, B – I, C – IV, D – II

(4) A – III B – IV, C – I, D – II

36. Which of the following is not a characteristic feature of the pBR322 vector?

(1) It is a *E.coli* cloning vector.

(2) It has antibiotic – resistance genes i.e., "tet^R" and "amp^R"

(3) The ampicillin resistance gene has restriction sites for "BamHI" and "SalI".

(4) The restriction site for "pvuII" is present in the "rop" sequence.

37. Read the following statements and choose the correct option.

Statement-I: The normal *E.coli* has resistance against antibiotics kanamycin and tetracyclin.

Statement-II: Selectable marker gene helps in the selection of transformants.

(1) Both Statement-I and Statement-II are true.

(2) Statement-I is false but Statement-II is true.

(3) Statement-I is true but Statement-II is false.

(4) Both Statement-I and Statement-II are false.

38. Which of the following statements is incorrect with respect to gene cloning?

(1)Cosmid: A hybrid molecule that combines features of plasmids and phages, used to clone larger DNA fragments.

(2) Plasmid: A linear, single-stranded DNA molecule used in genetic engineering.

(3) Lambda phage: A virus that infects bacteria and is often used for DNA cloning.

(4) Electroporation: A technique used to introduce foreign DNA into cells by applying an electrical field

39. The RNA interference (RNAi) technique has been successfully used to control ——— infection in tobacco plants:

(1) *Ascaris lumbricoides*

(2) *Pseudomonas*

(3) *Streptococcus pneumoniae*

(4) *Meloidogyne incognitia*

40. CryIIAb and CryIAb produce toxins that control.

(1) Dragonfly.

(2) Cotton bollworms and corn borer, respectively.

(3) Aphids.

(4) Nematodes and armyworms respectively.

41. The substrate concentration at which the chemical reaction catalyzed by an enzyme reaches half of its maximum velocity is called:

(1) Km or Michaelis constant

(2) ½ of Vmax

(3)Vmax constant

(4) Enzyme-substrate constant

42. Pith tissue is well developed in

(1) Monocot root and monocot stem

(2) Dicot root and dicot stem

(3) Monocot root and dicot stem

(4) Dicot root and monocot stem

43. Monoadelphous stamens seen in the members of

(1) Asteraceae

(2) Malvaceae

(3) Solanaceae

(4) Brassicaceae

44. If a single cell undergoes 7 generations of mitosis, calculate the total number of cells produced and the number of mitotic divisions required by using formulae given below.

Formula 1:

Total number of cells produced after n generations is given by $N = 2^n$

Formula 2:

The number of mitotic divisions required to produce N cells is N - 1

(1) 128 cells ; 127 divisions

(2) 64 cells; 63 divisions

(3) 128 cells; 64 divisions

(4) 256 cells; 255 divisions

45. Which of the following statements is false with respect to photorespiration?

(1) Oxygenation of RuBP occurs in the chloroplast.

(2) Neither sugars nor ATP are synthesized during photorespiration.

(3) Synthesis of NADPH occurs during photorespiration.

(4) One molecule of phosphoglycerate and one molecule of phosphoglycolate are formed per oxygenation of RuBP.

ZOOLOGY

46. Which of the following is not a characteristic of class chondrichthyes

(1) Gill slits are without operculum

(2) They are predaceous

(3) Air bladder is present

(4) Notochord is persistent throughout the life

47. In frog, bidder's canal is

(1) Found in kidney and transports sperms.

(2) Found in testis and transports sperms.

(3) Found in kidney and transports both urine and sperms.

(4) Found in testis and transports both urine and sperms.

48. Choose the odd one with respect to skull bones in an adult human

(1) Hyoid (2) Incus

(3) Ethmoid (4) Ischium

49. Which of the following is incorrect ?

(1) The DNA finger print of a child matches 50% with that of either parent.

(2) The DNA finger prints of identical twins are very much similar to each other.

(3) DNA fingerprinting helps in determining population and genetic diversities.

(4) DNA finger prints of a son and a daughter of same pair of parents match 100% with each other.

50. Choose the correct statements related to tissues

(I) Adipose tissue has semifluid ground substance.

(II) The main tissue that provides structural frame to the body is areolar tissue.

(III) Compound epithelium helps in secretion and absorption.

(1) I and III (2) I, II and III

(3) II only (4) I only

51. Assertion(A): Aldosterone is a steroid hormone which mainly acts at the renal tubules.

Reason(R): Aldosterone stimulates excretion of sodium and water at DCT.

(1) Both A and R are correct and R is the correct explanation of A

(2) Both A and R are correct but R is NOT the correct explanation of A

(3) A is correct but R is not correct

(4) A is not correct but R is correct

52. Visceral nervous system

(1) Is a part of somatic nervous system.

(2) Relays impulses from viscera to central nervous system.

(3) Relays impulses from central neural system to skeletal muscle.

(4) Relays impulses from central neural system to voluntary organs.

53. Match List-I with List-II

List-I		List-II	
A	Comb plates	I	Fasciola
B	Suckers	II	Sycon
C	Medusa	III	Pleurobrachia
D	Spicules	IV	Aurelia

(1) A – IV, B – I, C – II, D - III

(2) A – III, B – I, C – IV, D – II

(3) A – II, B – I, C – IV, D – III

(4) A – I, B – II, C – III, D – IV

54. During oogenesis, primary oocyte is found in

(1) Primary follicle only

(2) Primary follicle and Secondary follicle only

(3) Primary follicle, Secondary follicle and Tertiary follicle

(4) Graafian follicle

55. Statement-I: Feather like gills are present in between hump and mantle in molluscans.

Statement-II: Some molluscs are terrestrial.

(1) Both Statement I and Statement II are true

(2) Statement I is false but Statement II is true

(3) Statement I is true but Statement II is false

(4) Both Statement I and Statement II are false

56. Identify the incorrect combination from the following

(1) Periodic abstinence – Natural method

(2) Cervical caps – Barrier method

(3) Lippes loop – Hormonal IUD

(4) Saheli – OCPs

57. Neural centres that control eating and drinking are found in

(1) Hypothalamus

(2) Pons

(3) Limbic system

(4) Cerebellum

58. Choose the incorrect statements related to muscle fibre

(I) A band has both thick and thin filaments.

(II) Sarcomere has one 'I' band and two halves of 'A' bands.

(III) 'Z' line passes through centre of 'I' band.

(1) I and II (2) I and III

(3) III only (4) II only

59. Ejaculatory duct of male cockroach

(1) Situated in genital pouch

(2) Present in 4th to 6th segment

(3) Opens into male gonopore

(4) Encloses phallomeres

60. Assertion(A): Possibility of females with Down's syndrome is more than that of males.

Reason(R): Down's syndrome results due to the aneuploidy.

(1) Both A and R are correct and R is the correct explanation of A

(2) Both A and R are correct but R is NOT the correct explanation of A

(3) A is correct but R is not correct

(4) A is not correct but R is correct

61. Which of the following statement is not true

(1) Lysozyme is an antibacterial agent.

(2) Interferon is an antiviral protein.

(3) Heparin is an anticoagulant.

(4) Histamine is an antiallergic.

62. Smack is obtained from

(1) latex of papaver somniferum.

(2) Inflorescence of cannabis sativa

(3) Leaves of erythroxylum coca.

(4) Inflorescence of papaver somniferum.

63. Match List-I with List-II

List-I		List-II	
A	Proliferative	I	1st – 5th day
B	Secretory phase	II	6th – 13th day
C	Menstrual phase	III	15th – 28th day
D	Ovulation	IV	14th day

(1) A – III, B – II, C – I, D - IV

(2) A – II, B – III, C – IV, D – I

(3) A – I, B – II, C – III, D – IV

(4) A – II, B – III, C – I, D – IV

64. Volume of air that will remain in the lungs after normal inspiration

(1) TV +ERV (2) TV +IRV

(3) TV +FRC (4) TV +IRV +RV

65. Statement-I: Chorionic villi and uterine tissue becomes interdigitated to form placenta.

Statement-II: Formation of placenta between embryo and maternal body is called implantation.

(1) Both Statement I and Statement II are true

(2) Statement I is false but Statement II is true

(3) Statement I is true but Statement II is false

(4) Both Statement I and Statement II are false

66. Choose the correct sequence of evolution

(1) Thecodonts → therapsids → mammals

(2) Pelycosaurs → thecondonts → dinosaurs

(3) Tracheophyte ancestors → zosterophyllum → lycopods

(4) Chlorophyte ancestors → Rhynia type plants → tracheophyte ancestors

67. Statement-I: Gause's competitive exclusion principle is true if resources are limiting.

Statement-II: Paul Ehrlich's hypothesis explains the importance of key species of an ecosystem.

(1) Both Statement I and Statement II are true

(2) Statement I is false but Statement II is true

(3) Statement I is true but Statement II is false

(4) Both Statement I and Statement II are false

68. If 0.6 J of energy is present at the secondary carnivore of an ecosystem, energy found in plants in the form of gross primary productivity in that ecosystem is _____.

[If the rate of respiration in plants is 25% of GPP]

(1) 600 J (2) 800 J

(3) 200 J (4) 60 J

69. In humans the male accessory ducts does not include

(1) Seminiferous tubule (2) Epididymis

(3) Vas deferens (4) Vasa efferentia

70. Match List-I with List-II

List-I		List-II	
A	Glucocorticoids	I	Maintains corpus luteum
B	Gonadotropins	II	Increase rate of respiration
C	Catecholamines	III	Act on central neural system
D	Androgens	IV	Maintains kidney functioning

(1) A – IV, B – I, C – II, D - III

(2) A – II, B – III, C – IV, D – I

(3) A – II, B – I, C – IV, D – III

(4) A – IV, B – III, C – II, D – I

71. Read the following statements and choose the incorrect one.

(I) Trichomoniasis is an RTI and is caused by the virus Trichomonas.

(II) STDs like genital herpes is completely curable if detected early.

(III) Infertility can be caused if STIs are not treated early.

(1) (II) and (III) (2) (III) and (I)

(3) (I) and (II) (4) (II) only

72. During normal functioning of heart, tricuspid and bicuspid valves are pushed open during

(1) Atrial systole

(2) Joint diastole

(3) Ventricular systole

(4) Atrial and ventricular systole

73. Assertion(A): Sense of hearing is absent in calotes.

Reason(R): Calotes does not have external ear openings.

(1) Both A and R are correct and R is the correct explanation of A

(2) Both A and R are correct but R is NOT the correct explanation of A

(3) A is correct but R is not correct

(4) A is not correct but R is correct

74. In human, how much percent of the genome codes for proteins

(1) 99.9% (2) 50%

(3) Greater than ten percent

(4) Less than two percent

75. Statement-I: Many transgenic animals are designed to increase our understanding of how genes contribute to the development of disease.

Statement-II: The first transgenic cow , Rosie, produced α-1-antitrypsin , used in the treatment of emphysema.

(1) Both Statement I and Statement II are true

(2) Statement I is false but Statement II is true

(3) Statement I is true but Statement II is false

(4) Both Statement I and Statement II are false

76. Match List-I with List-II

List-I		List-II	
A	Pillar like cells	I	Cardiac muscle
B	Branching pattern	II	Erythrocyte
C	Fusiform	III	Smooth muscle fibre
D	Biconcave	IV	Columnar epithelium

(1) A – III, B – I, C – II, D - IV

(2) A – IV, B – III, C – I, D – II

(3) A – IV, B – I, C – III, D – II

(4) A – I, B – III, C – II, D – IV

77. Hormones that promote the development of mammary glands are

(1) Prolactin and leutinizing hormone.

(2) Estradiol and progesterone.

(3) Prolactin and oxytocin.

(4) Progesterone and relaxin.

78. Among the following sets of examples for convergent evolution, select the incorrect option.

(1) Flipper of whale and wing of bat.

(2) Sweet potato and potato.

(3) Eyes of octopus and that of mammals.

(4) Wings of bird and wings of butterfly.

79. Statement-I: Entamoeba, a protozoan parasite causes abdominal pain and constipation.

Statement-II: Wuchereria, an intestinal parasite spreads through the bite of female mosquito.

(1) Both Statement I and Statement II are true

(2) Statement I is false but Statement II is true

(3) Statement I is true but Statement II is false

(4) Both Statement I and Statement II are false

80. "In recent years, exsitu conservation has advanced beyond keeping threatened species in enclosures". Which of the following supports this statement?

(1) Zoological park

(2) Cryopreservation

(3) Sacred grove

(4) Biodiversity hot spot

81. Nicotine in tobacco stimulates

(1) Adrenal cortex thus raising blood pressure.

(2) Adrenal medulla thus raising the blood pressure.

(3) Adrenal cortex thus lowering blood pressure.

(4) Adrenal medulla thus lowering the blood pressure.

82. Choose the statements related to Darwinism

(I) Ability to adapt has genetic basis.

(II) Many life forms do not exist anymore due to extinctions in different periods of history.

(III) Variations are small and directional.

(1) I, II and III

(2) I and II only

(3) II and III only

(4) I and III only

83. The hepatic portal vein drains blood from

(1) Heart to liver

(2) Liver to intestine

(3) Intestine to liver

(4) Liver to heart

84. Assertion(A): Lymph nodes acts as the secondary lymphoid organs.

Reason(R): Lymphocytes develop into effector cells in these organs.

(1) Both A and R are correct and R is the correct explanation of A

(2) Both A and R are correct but R is NOT the correct explanation of A

(3) A is correct but R is not correct

(4) A is not correct but R is correct

85. Graves disease is caused due to

(1) Hyposecretion of thyroid hormones.

(2) Hyposecretion of adrenal gland hormones.

(3) Autoimmunity

(4) Allergy

86. Choose the correct statements

(I) Nile perch got extinct due to over exploitation.

(II) Communities with more number of species shows more year to year variations in biomass.

(III) Tropical environments are less seasonal than temperate environments.

(1) I only (2) III only

(3) I and II (4) I and III

87. Female bird is produced when a sperm with

(1) 'W' chromosome fertilizes ovum with 'Z'.

(2) 'Z' chromosome fertilizes ovum with 'Z'.

(3) 'W' chromosome fertilizes ovum with 'W'.

(4) 'Z' chromosome fertilizes ovum with 'W'.

88. Match List-I with List-II

List-I		List-II	
A	Cartilaginous joint	I	Between atlas and axis
B	Gliding joint	II	Between tarsals
C	Hinge joint	III	Between two coxal bones
D	Pivot joint	IV	Elbow joint

(1) A – II, B – III, C – IV, D - I

(2) A – IV, B – I, C – III, D – II

(3) A – III, B – II, C – IV, D – I

(4) A – II, B – I, C – III, D – IV

89. Embryological evidence for evolution was proposed by

(1) Miller

(2) Louis Pasteur

(3) Ernst Haeckel

(4) Alfred Wallace

90. Statement-I: Small amounts of urea enter the thin segment of the ascending limb of Henle's loop which is transported back to the interstitium by the collecting tubule.

Statement-II: The osmolarity of interstitial fluid in kidneys increases towards inner medulla.

(1) Both Statement I and Statement II are true

(2) Statement I is false but Statement II is true

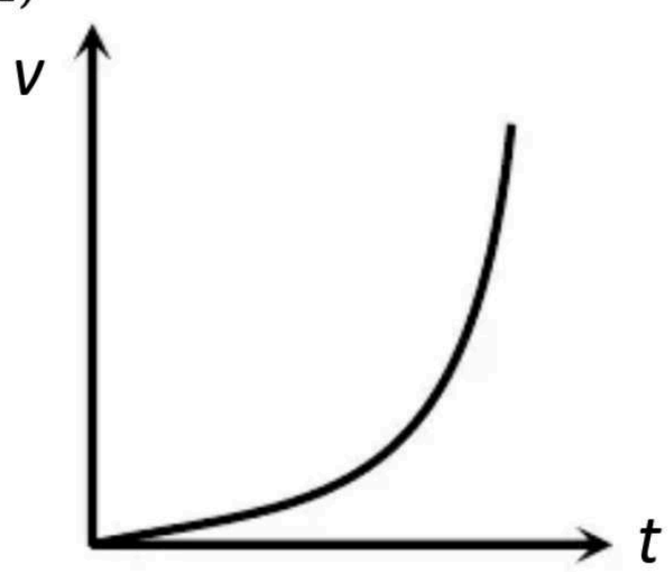
(3) Statement I is true but Statement II is false

(4) Both Statement I and Statement II are false

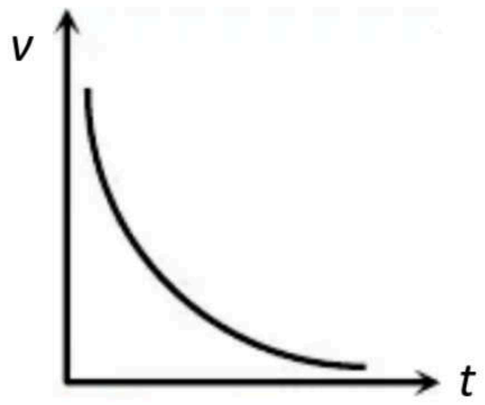
PHYSICS

91. Which of the following velocity-time graphs represent uniform motion.

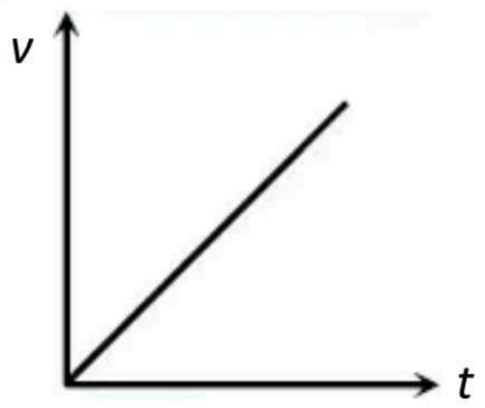
(1)



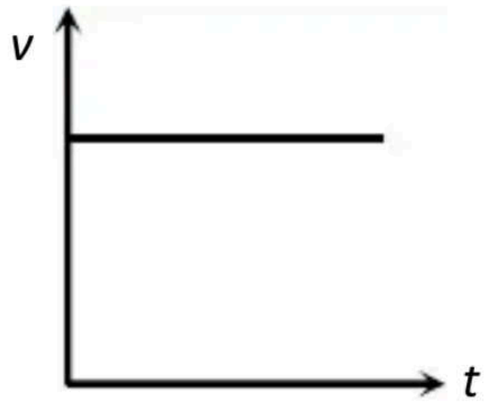
(2)



(3)



(4)



92. Which of the following algebraic operations with scalar and vector physical quantities are meaningful ?
A) adding any two scalars
B) adding any two vectors
C) multiplying a vector by a scalar
D) adding a component of a vector to the same vector
(1) only C
(2) A and B only
(3) A, B and C only
(4) A, B, C and D

93. The initial mass of a rocket is 1000 kg. Calculate at what rate the fuel should be burnt so that the rocket is given an acceleration of $20\,ms^{-2}$. The gases come out at a relative speed of $500\,ms^{-1}$ with respect to the rocket: (Use $g=10\,m/s^2$)
(1) $60\,kg\,s^{-1}$ (2) $10\,kg\,s^{-1}$
(3) $600\,kg\,s^{-1}$ (4) $500\,kg\,s^{-1}$

94. A block of mass 10 kg starts sliding on a surface with an initial velocity of $9.8\,ms^{-1}$. The coefficient of friction between the surface and block is 0.5 . The distance covered by the block before coming to rest is
(1) 4.9 m (2) 19.6 m
(3) 12.5 m (4) 9.8 m

95. A time dependent force $F=6t$ acts on a particle of mass 1 kg. If the particle starts from rest, the work done by the force during the first 1 second will be
(1) 4.5 J (2) 22 J
(3) 9 J (4) 18 J

96. Given below are two statements,one is labelled as Assertion A and the other is labelled as Reason R.
Assertion(A): The work done by the spring force in a cyclic process is zero.
Reason(R): The spring force is an example of a variable force which is non-conservative.
(1) Both A and R are correct and R is the correct explanation of A.
(2) Both A and R are correct and R is NOT the correct explanation of A.
(3) A is not correct but R is correct.
(4) A is correct but R is not correct.

97. The displacement of a particle executing simple harmonic motion is given by $y=A_0+A\sin\omega t+B\cos\omega t$. Then the amplitude of its oscillation is given by
(1) $A_0+\sqrt{A^2+B^2}$
(2) $\sqrt{A^2+B^2}$
(3) $\sqrt{A_0^2+(A+B)^2}$
(4) $A+B$

98. A simple pendulum attached to the roof of a lift has a time period of 2s in a stationary lift. If the lift is allowed to fall freely then the frequency of oscillations of pendulum will be
(1) zero (2) 2Hz
(3) 0.5 Hz (4) infinity

99. A force $\vec{F}=\alpha\hat{i}+3\hat{j}+6\hat{k}$ is acting at a point $\vec{r}=2\hat{i}-6\hat{j}-12\hat{k}$. The value of α for which angular momentum about origin is conserved is
(1) 2 (2) zero
(3) 1 (4) -1

100. Given below are two statements,one is labelled as Assertion A and the other is labelled as Reason R.
Assertion(A): Machines such as steam engine, the automobile engine, etc., that produce rotational motion have a disc with a large moment of inertia, called a flywheel.
Reason(R): Because of its large moment of inertia, the flywheel resists the sudden increase or decrease of the speed of the vehicle and prevents jerky motions, thereby ensuring a smooth ride.
(1) Both A and R are correct and R is the correct explanation of A.
(2) Both A and R are correct and R is NOT the correct explanation of A.
(3) A is not correct but R is correct.
(4) A is correct but R is not correct.

101. The fundamental frequency in an open organ pipe is equal to the third harmonic of a closed organ pipe. If the length of the closed organ pipe is 20 cm, the length of the open organ pipe is nearly
(1) 13.3 cm (2) 8 cm
(3) 16 cm (4) 12.5 cm

102. A string is stretched between two fixed points separated by 75.0 cm. It is observed to have resonant frequencies of 420Hz and 315 Hz. There are no other resonant frequencies between these two. The lowest resonant frequency for this string is
(1) 205 Hz (2) 10.5 Hz
(3) 105 Hz (4) 155 Hz

103. The radius of the convex surface of plano-convex lens is 20 cm and the refractive index of the material of the lens is 1.5. The focal length of the lens is
(1) 30 cm (2) 40 cm
(3) 20 cm (4) 50 cm

104. In total internal reflection , when the angle of incidence is equal to the critical angle for the pair of media in contact, what will be angle of refraction?
(1) 180^0 (2) 0^0
(3) equal to angle of incidence (4) 90^0

105. The two slits at a distance of 1 mm are illuminated by the light of wavelength $6.5\times10^{-7}m$. The interference fringes are observed on a screen placed at a distance of 1m. The distance between third dark fringe and fifth bright fringe will be
(1) 0.65 mm (2) 1.63 mm
(3) 3.25 mm (4) 4.88 mm

106. When the angle of incidence of a light ray from vacuum into a material is 60°, the reflected light is completely polarized. The velocity of the refracted ray inside the material is (in ms^{-1})
(1) 3×10^8 (2) $\left(\frac{3}{\sqrt{2}}\right)\times10^8$
(3) $\sqrt{3}\times10^8$ (4) 0.5×10^8

107. For the Bohr's first orbit of circumference $2\pi r$, the de-Broglie wavelength of revolving electron will be
(1) $2\pi r$ (2) πr
(3) $\frac{1}{2\pi r}$ (4) $\frac{1}{4\pi r}$

108. A photon of energy 3.4 eV is incident on a metal having work function 2 eV. The maximum K.E. of photo-electrons is equal to
(1) 6.8 eV (2) 1.7 eV
(3) 5.4 eV (4) 1.4 eV

109. In the spectrum of hydrogen atom, the ratio of the longest wavelength in Lyman series to the longest wavelength in the Balmer series is.
(1) 5/27 (2) 1/93
(3) 4/9 (4) 3/2

110. The energy liberated on complete fission of 1 kg of $_{92}U^{235}$ is (Assume 200 MeV energy is liberated on fission of 1 nucleus)
(1) $8.2\times10^{10}J$ (2) $8.2\times10^{9}J$
(3) $8.2\times10^{13}J$ (4) $8.2\times10^{16}J$

111. If a full wave rectifier circuit is operating from 50 Hz mains, the fundamental frequency in the ripple of output will be
(1) 50 Hz (2) 70.2 Hz
(3) 100 Hz (4) 25 Hz

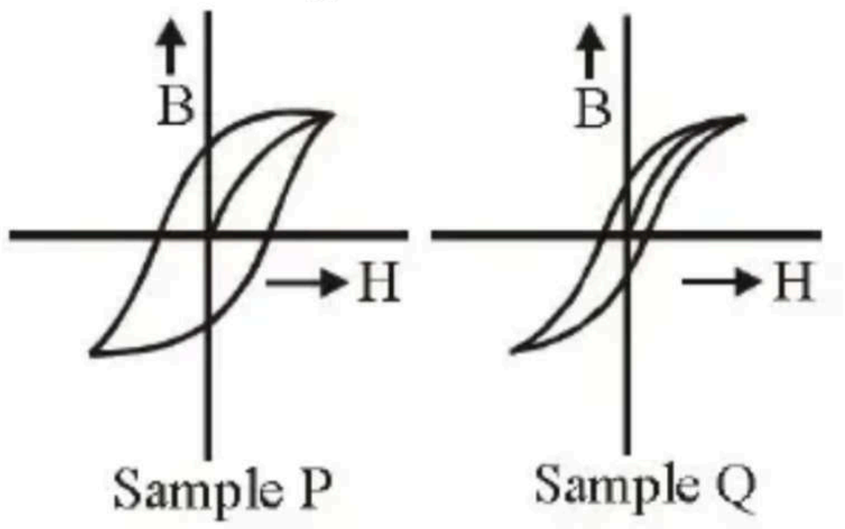
112. Given below are two statements .
STATEMENT(I): The I–V characteristics of solar cell is drawn in the fourth quadrant of the coordinate axes. This is because a solar cell does not draw current but supplies the same to the load.
STATEMENT(II): The reverse breakdown voltages of LEDs are very high. So care should be taken that high reverse voltages do not appear across them.
In the light of above statements, choose the correct answer from the options given below.
(1) both statement (I) and statement (II) are true.
(2) statement (I) is false but statement (II) is true
(3) statement (I) is true but statement (II) is false
(4) both statement (I) and statement (II) are false.

113. The force F is given in terms of time t and displacement x by the equation
$F = A\cos Bx + C\sin Dt$. The dimensional formula of $\frac{D}{B}$ is
(1) $[M^0L^0T^0]$ (2) $[M^0L^0T^{-1}]$
(3) $[M^0L^{-1}T^0]$ (4) $[M^0L^1T^{-1}]$

114. N division on the main scale of a vernier callipers coincide with (N + 1) divisions of the vernier scale. If each division of main scale is x unit, then least count of the instrument is
(1) $\frac{x}{N}$ (2) x
(3) $\left(\frac{N}{N+1}\right)x$ (4) $\frac{x}{N+1}$

115. What is the depth at which the value of acceleration due to gravity becomes $\frac{1}{n}$ times the value that at the surface of earth? (radius of earth is R)
(1) $\frac{R}{n}$ (2) $\frac{R}{n^2}$
(3) $\frac{R(n-1)}{n}$ (4) $\frac{Rn}{(n-1)}$

116. If the B-H curves of two samples of P and Q of iron are as shown below, then which one of the following statements is CORRECT ?

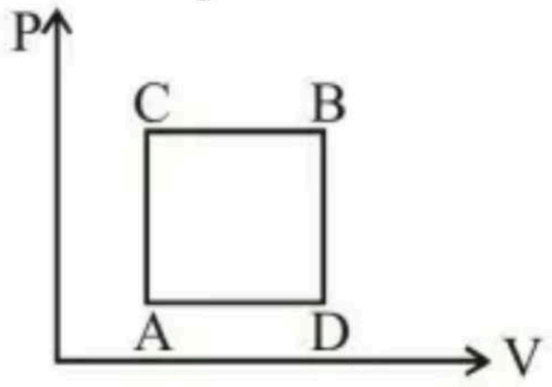


(1) Both P and Q are suitable for making permanent magnet
(2) P is suitable for making permanent magnet and Q for making electromagnet
(3) P is suitable for making electromagnet and Q is suitable for permanent magnet
(4) Both P and Q are suitable for making electromagnets

117. Assertion: A brass disc is just fitted in a hole in a steel plate. The system must be cooled to loosen the disc from the hole
Reason: The coefficient of liner expansion for brass is greater than the coefficient of linear expansion for steel.
(1) Both A and R are true, and R is the correct explanation of A
(2) Both A and R are true, and R is not the correct explanation of A
(3) A is true but R is false
(4) Both A and R are false

118. Two liquids A and B are at 32^0C and 24^0C. When mixed in equal masses, the temperature of mixture is found to be 28^0C. The ratio of specific heats of A and B is.
(1) 3:2 (2) 2:3
(3) 1:1 (4) 4:3

119. A gas can be taken from A to B via two different processes ACB and ADB.



When path ACB is used 60J of heat flows into the system and 30J of work is done by the system. If path ADB is used work done by the system is 10J. The heat flow into the system in path ADB is:
(1) 80J (2) 20J
(3) 100J (4) 40J

120. STATEMENT(A): The Zeroth Law of Thermodynamics led us to the concept of temperature
STATEMENT(B): The first law of thermodynamics is the general law of conservation of energy applied to any system in which energy transfer from or to the surroundings (through heat and work) is taken into account.
STATEMENT(C): The Second Law of Thermodynamics says that efficiency of a heat engine can never be unity. For a refrigerator, the Second Law says that the co-efficient of performance can never be infinite.
(1) Only A and B are true
(2) Only B and C are true
(3) Only A and C are true
(4) A, B and C all are true

121. The value of $\gamma\left(=\frac{C_p}{C_v}\right)$, for hydrogen, and helium are respectively equal to
(1) $\frac{7}{5},\frac{5}{3}$ (2) $\frac{5}{3},\frac{7}{5}$
(3) $\frac{3}{5},\frac{5}{7}$ (4) $\frac{9}{7},\frac{7}{5}$

122. Two wires are made of the same material and have the same volume. However, wire 1 has cross-sectional area A and wire 2 has cross-sectional area 3A. If the length of wire 1 increases by Δx on applying force F, how much force is needed to stretch wire 2 by the same amount?
(1) F (2) 2F
(3) 6F (4) 9F

123. A solid glass rod of radius 'r' is placed inside a glass cylinder of internal radius 'R'. This glass rod is placed coaxially with the cylinder and the system is held vertically with a part inside the water. Assuming angle of contact 0° and surface tension 'T' then the height to which water will rise in the region between the rod and cylinder is
(1) $\frac{2T}{(R-r)\rho g}$
(2) $\frac{T}{(R+r)\rho g}$
(3) $\frac{2T}{(R^2-r^2)\rho g}$
(4) $\frac{2T}{(R^2+r^2)\rho g}$

124. The condition under which a microwave oven heats up a food item containing water molecules most efficiently is
(1) The frequency of the microwaves has no relation with natural frequency of water molecules
(2) Microwave are heat waves, so always produce heating
(3) Infra-red waves produce heating in a microwave oven
(4) The frequency of the microwaves must match the resonant frequency of the water molecules

125. Match Lits-I with List-II

List-I:

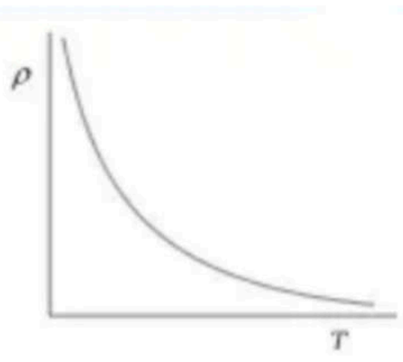
A) Resistivity ρ_T of copper as a function of temperature T.

B) Resistivity ρ_T of nichrome as a function of temperature T.

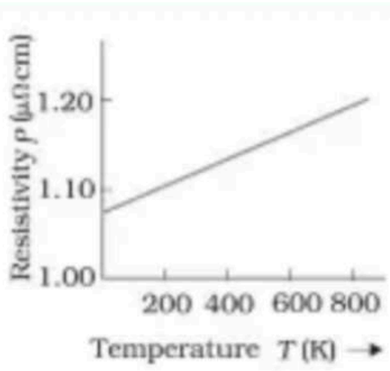
C) Temperature dependence of resistivity for a typical semiconductor.

List-II:

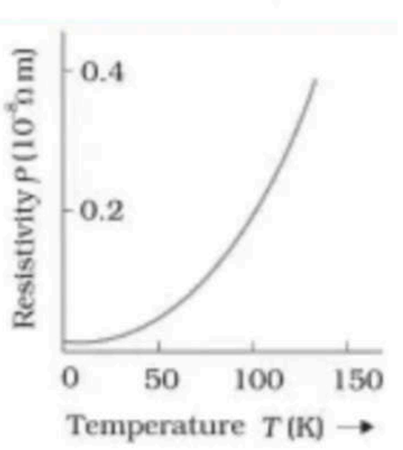
I)



II)

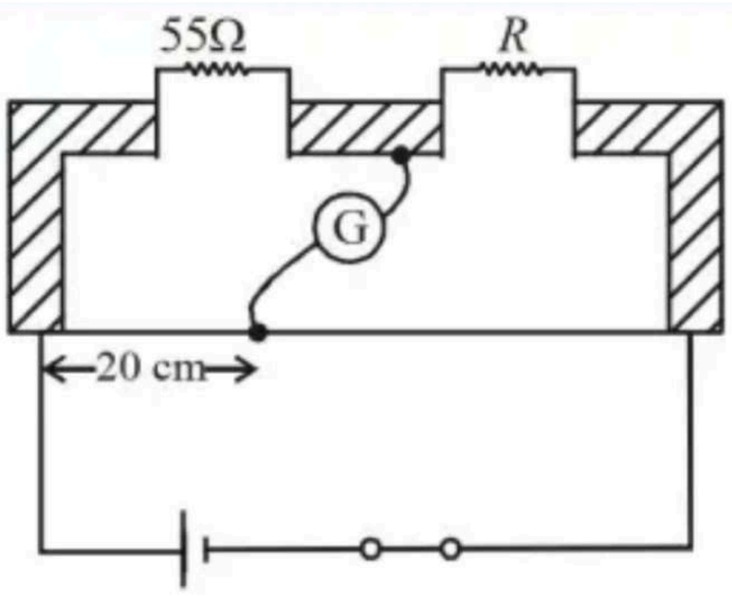


III)



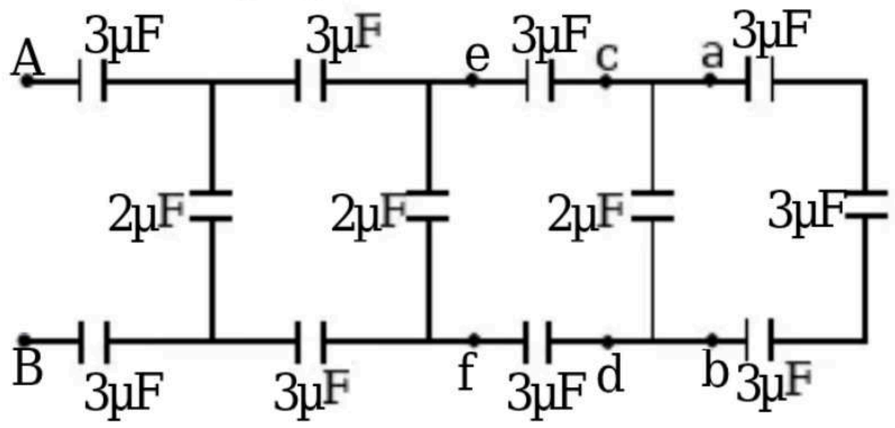
(1) (A)-II, (B)-III, (C)-I
(2) (A)-III, (B)-I, (C)-II
(3) (A)-I, (B)-II, (C)-III
(4) (A)-III, (B)-II, (C)-I

126. Shown in the figure below is a meter-bridge set up with null deflection in the galvanometer. The value of the unknown resistor R is



(1) 13.75Ω (2) 110Ω
(3) 220Ω (4) 55Ω

127. The resultant capacitance between A and B in the following figure is –



(1) 1μF (2) 10μF
(3) 50μF (4) 1.5μF

128. An electric dipole of moment p is placed in an electric field of intensity E. The dipole acquires a position such that the axis of the dipole makes an angle θ with the direction of the field. Assuming that the potential energy of the dipole to be zero when $\theta = 90^0$, the torque and the potential energy of the dipole will respectively be

(1) $pE\cos\theta, -pE\sin\theta$
(2) $pE\sin\theta, -2pE\cos\theta$
(3) $pE\sin\theta, 2pE\cos\theta$
(4) $pE\sin\theta, -pE\cos\theta$

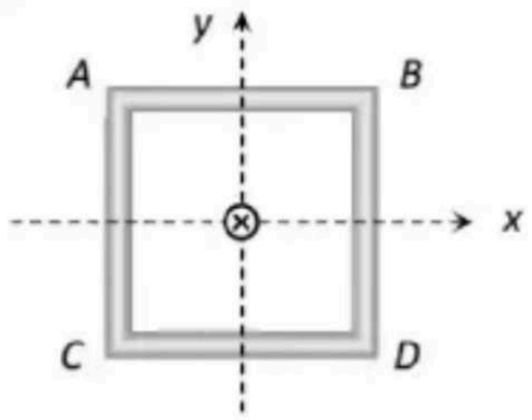
129. An imaginary closed spherical surface S of radius R is cantered on the origin. A positive charge +q is originally at the origin and electric flux through the surface is φ_E. Three additional charges are now added along the x axis: $-3q$ at $x = -\frac{R}{2}$, $+5q$ at $x = +\frac{R}{2}$ and $4q$ at $x = +\frac{3R}{2}$ then flux through S is now

(1) $6\varphi_E$ (2) $7\varphi_E$
(3) $3\varphi_E$ (4) $8\varphi_E$

130. A charge 'q' is placed at the centre of the line joining two equal charges 'Q'. The system of the three charges will be in equilibrium if 'q' is equal to

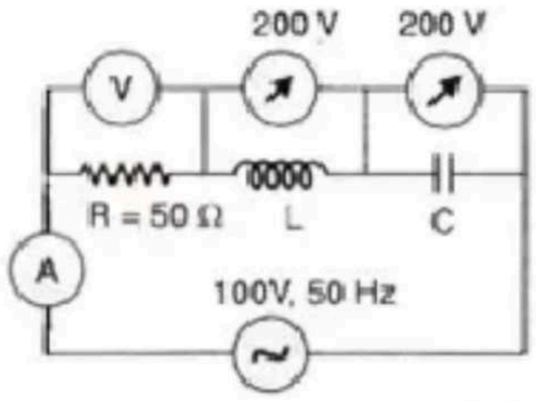
(1) Q/2 (2) – Q/4
(3) Q/4 (4) – Q/2

131. A square coil $ABCD$ lying in x-y plane with it's centre at origin. A long straight wire passing through origin carries a current $i = 2t$ in negative z-direction. The induced current in the coil is



(1) Clockwise (2) Anticlockwise
(3) Alternating (4) Zero

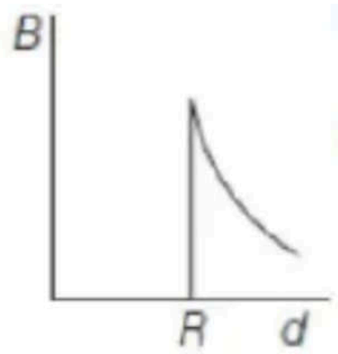
132. In the series LCR circuit, the voltmeter and ammeter readings are respectively:



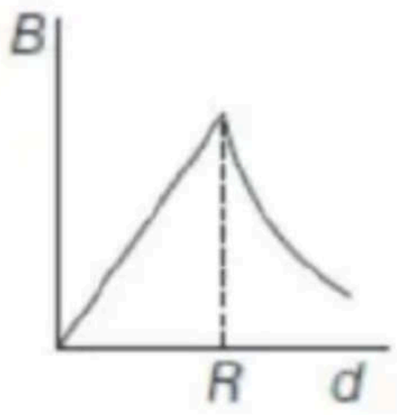
(1) V = 250V, I = 4 A
(2) V = 150V, I = 2 A
(3) V = 1000V, I = 5 A
(4) V = 100V, I = 2 A

133. A cylindrical conductor of radius R is carrying a constant current. The plot of the magnitude of the magnetic field B with the distance d from the centre of the conductor, is correctly represented by the figure.

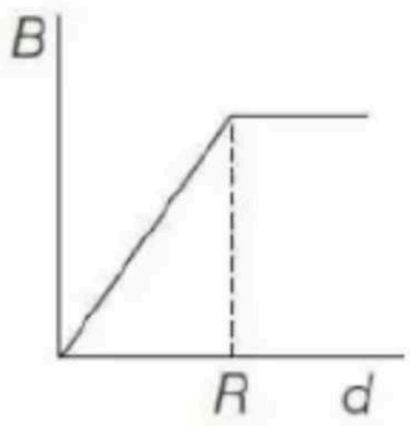
(1)



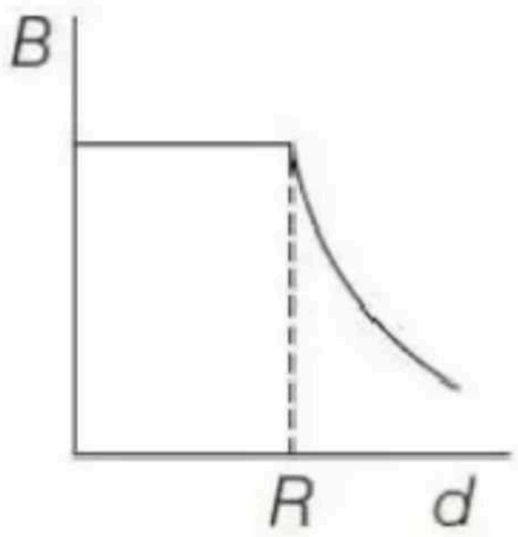
(2)



(3)



(4)



134. STATEMENT(A) : Increasing the current sensitivity of moving coil galvanometer may not necessarily increase its voltage sensitivity
STATEMENT(B) : A charged particle moves through a magnetic field perpendicular to its direction. Then its momentum changes but the kinetic energy is constant
(1) A is true, B is false
(2) A is false, B is true
(3) Both A and B are true
(4) Both A and B are false

135. A circuit to verify Ohms law uses ammeter and voltmeter in series or parallel connected correctly to the resistor. In the circuit
(1) Ammeter is always used in parallel and voltmeter in series
(2) Both ammeter and voltmeter must be connected in parallel
(3) Ammeter is always used in series and voltmeter in parallel
(4) Both ammeter and voltmeter must be connected in series

CHEMISTRY

136. Number of neutrons of 14 mg of C^{14} isotope
(1) 4.2×10^{23} (2) 6.0×10^{20}
(3) 6.0×10^{23} (4) 48×10^{20}

137. Line spectrum is characteristic of
(1) Atoms
(2) Molecules
(3) Any substance in the solid state
(4) Both atoms and molecules

138. de – Broglie wave length is maximum for which of the following particle travelling with same velocity
(1) Electron (2) Alpha particle
(3) Sodium atom (4) Proton

139. Number of significant figures in 202051 is
(1) 6 (2) 4
(3) 3 (4) 2

140. 1 amu is equal to
(1) 0.01 gm (2) 0.01 mg
(3) 0.01 kg (4) 1.66×10^{-24} g

141. Assertion (A) : Bond order in O_2^+ is greater than that of O_2^-
Reason (R) : O_2^+ is para magnetic while O_2^- is dia magnetic
(1) Both A and R are true and the R is the correct explanation of the A.
(2) Both A and R are true but the R is not the correct explanation of the A.
(3) A is true but R is false.
(4) Both A and R are false

142. Hybridisation of Boron in BF_3 molecule is
(1) sp^3 (2) sp^2
(3) sp (4) sp^3d

143. Statement – I : When a system is in equilibrium, the entropy is maximum and the change in entropy is zero.
Statement – II : When a system is in equilibrium the free energy of system is minimum and change in free energy is equal to zero.
(1) Both S - I and S - II are correct
(2) Both S - I and S - II are incorrect
(3) S - I is correct but S - II is incorrect
(4) S - I is incorrect but S - II is correct

144. The following reaction is accompanied by an increase in enthaply
(1) Formation of H_2O from H_2 & O_2
(2) Formation of NO form N_2 & O_2
(3) Formation of CO_2 from C & O_2
(4) Formation of SO_2 from S & O_2

145. Select the correct statements
a) k_{sp} is influenced by common ion effect
b) k_{sp} is influenced by temperature
c) Solubility (s) is influenced by temperature
(1) All are correct (2) Only b, c
(3) Only a, c (4) Only a, b

146. p^H of a buffer solution containing equimolar ratio of weak acid H A and its conjugate base A^- will be
(1) p^{ka} (2) $p^{ka}+1$
(3) $p^{ka}-1$ (4) $p^{ka}+0.5$

147. 4 mole of PCl_5 is taken in a 1L vessel and heated to 273°C to establish $\left[PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}\right]$. At equilibrium 2 mole of Cl_2 is formed. The value of k_c at this temperature is
(1) $\frac{1}{3}$ mole / L (2) 0.5 mole / L
(3) 2 mole / L (4) 1 mole / L

148. For a gas phase reaction, the value of $k_c = 5$ mole / L. If pressure is increased, then at the same temperature
(1) For ward reaction is favoured
(2) Backward reaction is favoured
(3) Reaction is uneffected
(4) k_c value increases

149. The electrode potential of hydrogen electrode will be maximum if the solution used is
(1) 0.1 M HCl
(2) 0.1 M H_2SO_4
(3) 0.1 M CH_3COOH
(4) 0.1 M NaOH

150. For a cell reaction Mg / Mg^{+2} (xM) / F^- (yM) / F_2 , Pt. The E.M.F will be maximum if x and y respectively are
(1) 1M, 1M
(2) 1M, 0.01 M
(3) 0.01 M, 0.01 M
(4) 0.01M, 1M

151. Units for cell constant (G^*) are
(1) cm (2) cm^2
(3) m^2 (4) m^{-1}

152. For the reaction $2SO_2(g) + O_2(g) \longrightarrow 2SO_3(g)$. If the rate of disappearance of SO_2 is 32 mg/sec, then the rate of disappearance of O_2 will be
(1) 64 mg/sec (2) 32 mg/sec
(3) 16 mg/sec (4) 8 mg/sec

153. Units for rate constant for 1st order reaction are
(1) sec^{-1}
(2) $M.sec^{-1}$
(3) $L.\ mole^{-1}\ sec^{-1}$
(4) $L^2\ mole^{-2}\ sec^{-1}$

154. Assertion (A) : Vant Hoff's factor 'i' increases as conc. of the electrolytic solution increases
Reason (R) : At infinite dilution 'i' for $MgCl_2$ is 3
(1) Both A and R are true and the R is the correct explanation of the A.
(2) Both A and R are true but the R is not the correct explanation of the A.
(3) A is true but R is false.
(4) A is false and R are true

155. 3 mole of a non electrolyte is dissolved in 97 mole of water. Then R L V P of the solution
(1) 0.05 (2) 0.04
(3) 0.03 (4) 0.01

156. The linkage that joins two nucleotide in a poly a nucleotide is
(1) N – Glycoside
(2) Phospho ester bond
(3) Phospho diester bond
(4) O – gly coside

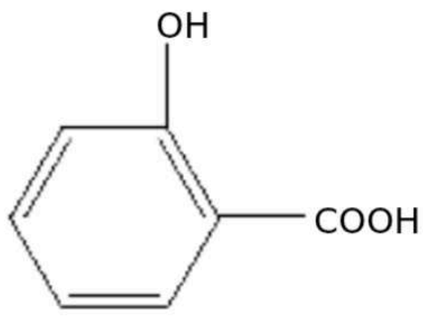
157. Non reducing sugar among the following
(1) Sucrose (2) Maltose
(3) Fructose (4) Lactose

158. Identify the incorrect match.

Name		IUPAC official name	
a)	Unniluniium	i)	Mendelevium
b)	Unniltrium	ii)	Lawrencium
c)	Unnilhexium	iii)	Seaborgium
d)	Ununumnium	iv)	Darmstadtium

(1) (d), (iv) (2) (a), (i)
(3) (b), (ii) (4) (c), (iii)

159. Match Column I with Column II and choose the correct combination from the options given.

Column I (reaction)		Column II (name of the reactions)	
A	$C_6H_5CHO + NaOH \rightarrow C_6H_5COOH + C_6H_5CH_2OH$	p	Rosenmund's reaction
B	$C_6H_6 + CH_3Cl \xrightarrow[AlCl_3]{Anhyd} C_6H_5CH_3$	q	Kolbe's reaction
C	$RCOCl + H_2 \xrightarrow{Pd-BaSO_4} RCHO$	r.	Friedel – Crafts reaction
D	$C_6H_5OH + NaOH + CO_2 \xrightarrow[H^+]{Heat}$ 	S	Cannizzaro's reaction

(1) (A) – s; (B) – r; (C) – p; (D) – q
(2) (A) – q; (B) – r; (C) – p; (D) – s
(3) (A) – p; (B) – q; (C) – r; (D) – s
(4) (A) – r; (B) – s; (C) – q; (D) – p

160. The incorrect statement regarding chirality is
(1) Enatiomers are superimposable mirror images of each other.
(2) A racemic mixture shows zero optical rotation.
(3) S_N1 reaction yields 1 : 1 mixture of both enantiomers
(4) The product obtained by S_N2 reaction of haloalkane having chirality at the reactive site shows inversion of configuration

161. Which of the following is the correct order of increasing field strength of ligands to form coordination compounds?

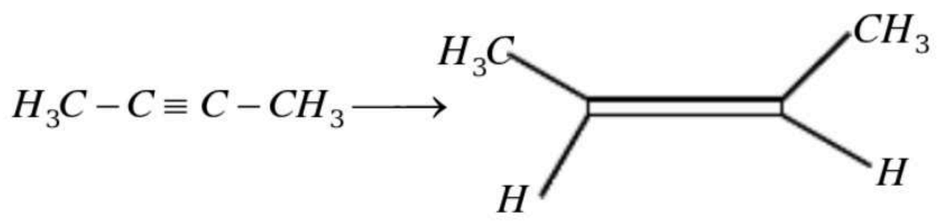
(1) $I^- < Br^- < F^- < Cl^-$

(2) $I^- < Br^- < Cl^- < F^-$

(3) $F^- < Br^- < Cl^- < I^-$

(4) $F^- < Cl^- < Br^- < I^-$

162. The most suitable reagent for the following conversion, is



(1) $Hg^{2+} / H^+, H_2O$

(2) $Na / liquid\ NH_3$

(3) $H_2, Pd / C, quinoline$

(4) Zn / HCl

163. Given below are two statements:

Statement I : Controlled nitration of aniline can be done conveniently by protecting $-NH_2$ group through acetylation.

Statement II : Direct nitration of aniline yields ortho,meta and para derivatives.

In the light of above statements,choose the most appropriate answer from the options given below:

(1) Both statement - I & II are true

(2) Statement – I is true and statement – II is false

(3) Both statement – I & II are false

(4) Statement – I is false and statement II is true

164. The transition element that has lowest enthalpy of atomization is:

(1) Fe (2) Cu

(3) Co (4) Sc

165. Permanganometric titrations in presence of HCl are not satisfactory because

(1) HCl is a weak acid

(2) HCl is a volatile acid

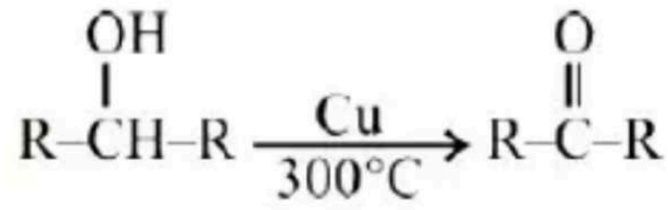
(3) $KMnO_4$ will oxidise HCl into Cl_2

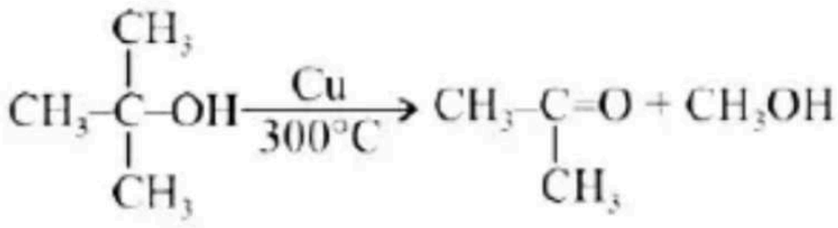
(4) $KMnO_4$ will show disproportionation in presence of HCl

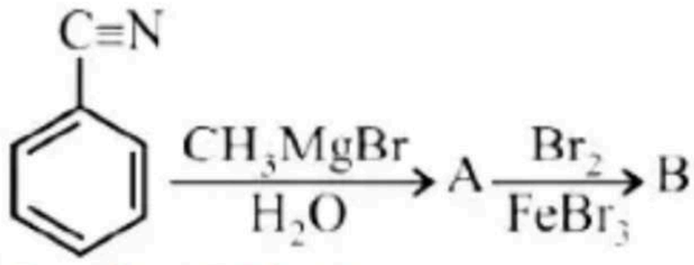
166. Which of the following is incorrect?

(1) $R-CH_2-OH \xrightarrow{P.C.C} R-CH=O$

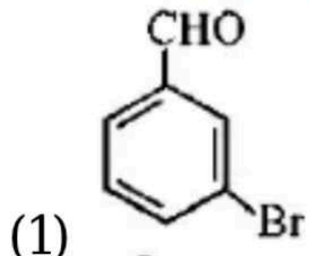
(2) $R-CH_2-OH \xrightarrow[300^oC]{Cu} R-CH=O$

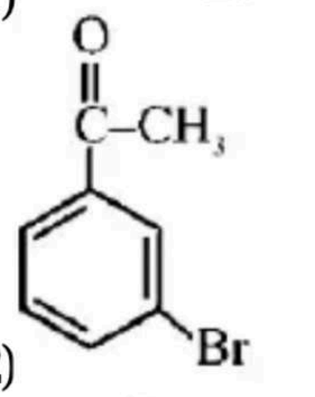
(3) 

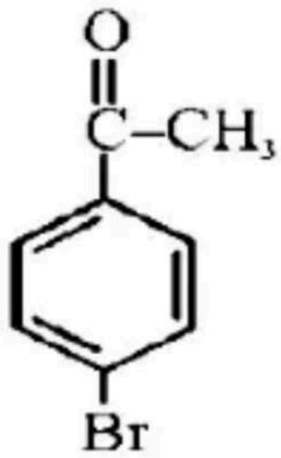
(4) 

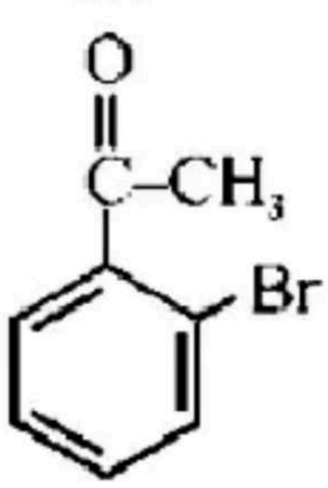
167. 

Product B is :-

(1) 

(2) 

(3) 

(4) 

168. Acetaldehyde reacts with semicrabazide, product will be

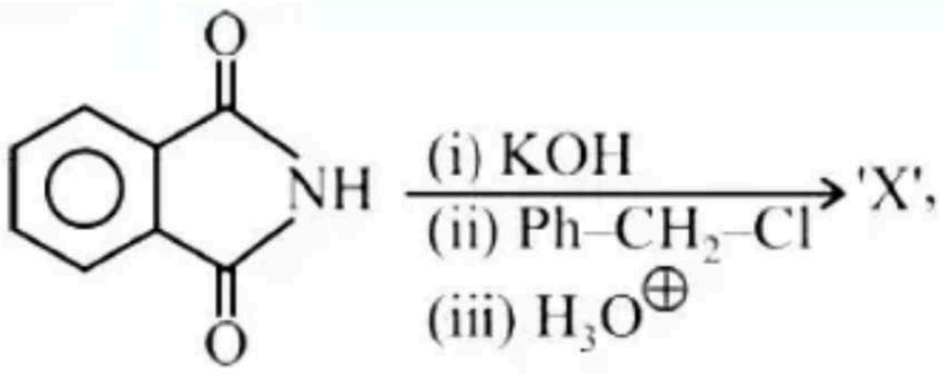
(1) $CH_3CH=N-NH-\overset{\overset{O}{\|}}{C}-NH_2$

(2) $CH_3CHNCONHNH_2$

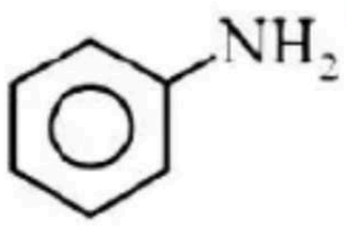
(3) $CH_3CHNHNH_2$

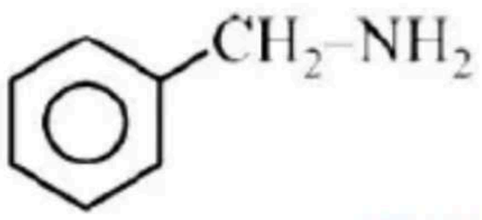
(4) $CH_3-\overset{\overset{O}{\|}}{C}-NH-CONH_2$

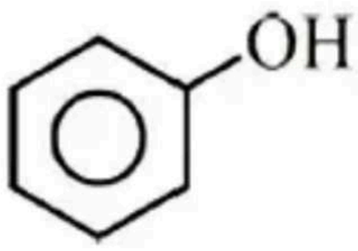
169.

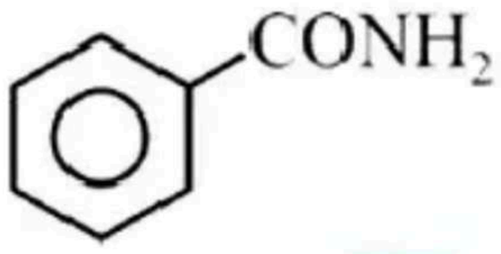


Product 'X' is:

(1) 

(2) 

(3) 

(4) 

170. Which of the following haloalkanes reacts with aqueous KOH most easily (SN^1 mechanism)?
(1) 1-Bromobutane
(2) 2-Bromobutane
(3) 2-Bromo-2-methylbutane
(4) 1-Chlorobutane

171. Methods used for purification of organic compounds are based on:
(1) Neither on nature of compound nor on the impurity present
(2) Nature of compound only
(3) Nature of compound and presence of impurity
(4) Presence of impurity only

172. Most basic among the given below
(1) Aniline
(2) Dimethyl amine
(3) Ammonia
(4) Pyrole

173. Match List I with List II

List – I		List – II	
A.	$K_2[Ni(CN)_4]$	I.	sp^3
B.	$[Ni(CO)_4]$	II.	sp^3d^2
C.	$[Co(NH_3)_6]Cl_3$	III.	dsp^2
D.	$Na_3[CoF_6]$	IV.	d^2sp^3

Choose the correct answer from the options given below:
(1) A – III, B – I, C – II, D – IV
(2) A – III, B – II, C – IV, D – I
(3) A – I, B – III, C – II, D – IV
(4) A – III, B – I, C – IV, D – II

174. Correct increasing order of atomic radius is represented in
(1) B <Al <Ga <In <Tl
(2) B <Ga <In <Al <Tl
(3) B <Ga <Tl <In <Al
(4) B <Ga <Al <In <Tl

175. Assertion(A): $C_6H_5-O-CH_3$ when heated with conc. HI gives C_6H_5-OH and CH_3I.
Reason(R): Reaction of $C_6H_5-O-CH_3$ and conc. HI/Δ Proceeds by SN^1 mechanism.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

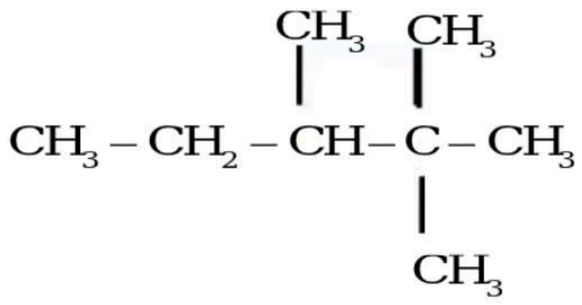
176. IUPAC group prefix name of -CHO is
(1) aldehyde
(2) oxo
(3) al
(4) carbaldehyde

177. Phenol is brominated in solvent CS_2 at low temperature. The product formed are:
(1) Monobromophenols
(2) Dibromophenols
(3) Tribromophenols
(4) Tribromobenzene

178. Which one of the following order is not in accordance with the property stated against it?
(1) F >Cl >Br >I: Elector negativity
(2) $F_2 > Cl_2 > Br_2 > I_2$: Bond dissociation energy
(3) $F_2 > Cl_2 > Br_2 > I_2$: Oxidizing power
(4) HI >HBr >HCl >HF: Acidic strength

179. Temporary electron displacement effect is
(1) Inductive effect
(2) Electromeric effect
(3) Resonance effect
(4) Hyper conjugation

180. Number of primary, secondary, tertiary and neo carbons in the following compound respectively are



(1) 4, 2, 1, 1
(2) 5, 1, 1, 1
(3) 3, 2, 2, 1
(4) 5, 0, 2, 1